

Object Oriented Programming using Java

Course Information

Degree: UG

Program: BCA

Course Name: Object Oriented Programming using Java

Course Credits: 2

Semester: 2

Course Category: Core

Type of Course: Practical / Experiential

New Course: No

Learning Hours: 30

Program Outcomes:

PO1 (Foundation Knowledge): Apply mathematical concepts and programming fundamentals to develop solutions for IT systems and applications.

PO2 (Problem Analysis): Analyze and interpret IT requirements to identify, formulate, and solve computational problems using standard practices.

PO3 (Design Development of Solutions): Design and develop IT solutions, including software and applications, to address real-world challenges.

PO4 (Modern Tool Usage): Use modern Computing tools and techniques necessary for innovative technological solutions.

PO5 (Individual and Teamwork): Work effectively both as an individual and as part of a team in multidisciplinary IT projects.

PO6 (Project Management): Apply project management principles and agile methodologies to plan, schedule, and execute IT projects.

PO7 (Ethics and Social Responsibility): Demonstrate ethical practices in IT solutions, including data privacy, cybersecurity, and sustainable development.

PO8 (Life-long Learning): Engage in continuous learning through certifications, workshops, and industry-relevant training to adapt to emerging technologies.

PO9 (Entrepreneurship and Innovation): Demonstrate innovation and entrepreneurial skills by identifying opportunities to create IT solutions for business and societal needs.

Course Outcomes:

CO#	Course Outcome	Bloom's Taxonomy Level
CO1	Understand the basics of Java programming	BL-2
CO2	Apply Object-Oriented concepts	BL-3
CO3	Implement exception handling and multithreading	BL-4
CO4	Work with GUI components & event handling	BL-4
CO5	Develop applications using Java	BL-5

Course Content Breakdown (Modules - 30 Hours)

Module	Course Content	Tools Used / Assessment and Activity	Bloom's Level	Mapping COs	Percentage change in Module
Module 1	Introduction to Java (6 Hours)	Hands-on coding exercises. Think-Analyze – Present Hacker-rank 100 Days Challenges			NEW
	Overview of Java & Features		BL-2	C01	
	JVM, JRE, JDK		BL-2	C01	
	Java Program Structure & Compilation		BL-3	C01	
	Data Types, Variables, Operators		BL-2	C01	
	Control Statements (if-else, loops)		BL-3	C01	
Module 2	Object-Oriented Programming in Java (6 Hours)				NEW
	Classes and Objects	Hacker-rank 100 Days Challenges	BL-3	C02	
	Constructors and Method Overloading	Mini-Project	BL-3	C02	

Module	Course Content	Tools Used / Assessment and Activity	Bloom's Level	Mapping COs	Percentage change in Module
	Inheritance and Polymorphism		BL-4	C02	
	Abstract Classes and Interfaces		BL-4	C02	
Module 3	Exception Handling & Multithreading (6 Hours)				NEW
	Try-catch-finally, Throw & Throws	Hacker-rank 100 Days Challenges Mini-Project	BL-3	C03	
	Custom Exceptions		BL-4	C03	
	Threads and Runnable Interface		BL-4	C03	
	Synchronization		BL-4	C03	
Module 4	Java I/O & Collections Framework (6 Hours)				NEW
	File Handling (Reading & Writing)	Hacker-rank 100 Days Challenges Mini-Project Pre-Final-Assessment	BL-3	C04	
	Serialization & Deserialization		BL-4	C04	
	ArrayList, LinkedList, HashMap		BL-3	C04	
	Iterators & Lambda Expressions		BL-4	C04	
Module 5	GUI Programming & Database Connectivity (6 Hours)				NEW
	Swing and AWT Components	Hacker-rank 100 Days Challenges	BL-3	C05	

Module	Course Content	Tools Used / Assessment and Activity	Bloom's Level	Mapping COs	Percentage change in Module
	Event Handling (Listeners)	Mini-Project	BL-4	C05	
	JDBC Connectivity (CRUD Operations)		BL-4	C05	
	Mini Project using Java		BL-5	C05	

Learning Hours

Activity	Hours per Week
Lecture	2
Tutorial	0
Practical	2
Experiential Learning	1

Course Contribution

Category	Percentage Contribution
Employment Enhancement	30%
Entrepreneurship Skills	10%
Skill Development	60%

Course Relevance

Level	Percentage Relevance
Global	20%
National	30%
Regional	30%
Local	20%

CO-PO/PSO Mapping

CO	P01	P02	P03	P04	P05	P06	P07	P08	P09	PS01	PS02
C01	3	1	1	2	-	-	-	2	-	2	2
C02	3	2	2	2	1	-	-	2	-	2	3
C03	3	2	2	2	1	1	-	2	-	2	2
C04	3	2	2	2	1	1	-	2	-	2	2
C05	3	3	3	2	3	3	1	3	2	3	3